Algebra 1 | Name____________________
Unit 6 | Date____________________
Lesson 2 Practice | Period____________________

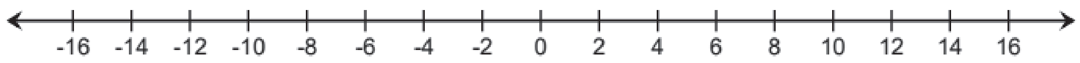
Use the following conjunctive inequality to answer questions 1-3.

$$-24 - 2x \leq -\frac{x}{3} - 4 < -2x + 16$$

Rewrite the compound inequality as two single inequalities and solve each one.

1.	2.

3. Graph the solution set of the compound inequality.



Alex just started a new sales floor job to save for college. He earns $\$16.25$ per hour plus a flat fee of $\$55$ per week. He wants to earn between $\$250$ and $\$445$ per week. The following inequality represents his earning potential:

$$250 \leq 16.25x + 55 \leq 445$$

4. Solve the inequality.

5. Graph the solution on the number line.

6. Complete the statement: Alex can work between ______ and ______ hours each week.

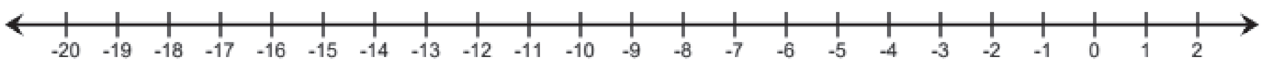
Use the disjunctive inequality shown below to answer questions 7-10:

$$-\frac{5}{3}\left(\frac{4}{3}k+\frac{8}{5}\right) > -3\frac{1}{4}k-\frac{53}{6} \quad \text{or} \quad \frac{24k+14}{5} \leq -74$$

Solve each component of the compound inequality.

7. $-\frac{5}{3}\left(\frac{4}{3}k+\frac{8}{5}\right) > -3\frac{1}{4}k-\frac{53}{6}$	8. $\frac{24k+14}{5} \leq -74$

9. Graph the solution sets of both inequalities on the number line.



10. Explain which of the following values is not a solution to the inequality.

-19	-9	1